

Runir's Description Logic Feature Syntax and Semantics

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1 Preliminaries

Let $\mathcal{I}$ be an interpretation with finite, possibly empty domain $\Delta^{\mathcal{I}}$. Each atomic concept A is interpreted as a set $(A)^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$, each atomic role R as a binary relation $(R)^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$, and each individual name a as an element $(a)^{\mathcal{I}} \in \Delta^{\mathcal{I}}$.

Concept descriptions denote subsets of $\Delta^{\mathcal{I}}$. Role descriptions denote binary relations over $\Delta^{\mathcal{I}}$. For planning states, let s denote the set of atoms true in the current state, and let γ denote the set of goal literals.

2 Concept Constructors

Name	Syntax	Semantics
Top	$\top$	$(\top)^{\mathcal{I}} = \Delta^{\mathcal{I}}$
Bottom	$\perp$	$(\perp)^{\mathcal{I}} = \emptyset$
Atomic state concept	P	$(P)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid P(a) \in s\}$
Positive atomic goal concept	G^+	$(G^+)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid G(a) \in \gamma\}$
Negative atomic goal concept	G^-	$(G^-)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \neg G(a) \in \gamma\}$
Intersection	$C \sqcap D$	$(C \sqcap D)^{\mathcal{I}} = (C)^{\mathcal{I}} \cap (D)^{\mathcal{I}}$
Union	$C \sqcup D$	$(C \sqcup D)^{\mathcal{I}} = (C)^{\mathcal{I}} \cup (D)^{\mathcal{I}}$
Negation	$\neg C$	$(\neg C)^{\mathcal{I}} = \Delta^{\mathcal{I}} \setminus (C)^{\mathcal{I}}$
Value restriction	$\forall R.C$	$(\forall R.C)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \forall b. (a, b) \in (R)^{\mathcal{I}} \Rightarrow b \in (C)^{\mathcal{I}}\}$
Existential quantification	$\exists R.C$	$(\exists R.C)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \exists b. (a, b) \in (R)^{\mathcal{I}} \wedge b \in (C)^{\mathcal{I}}\}$
At-least number restriction	$\geq n R$	$(\geq n R)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \{b \in \Delta^{\mathcal{I}} \mid (a, b) \in (R)^{\mathcal{I}}\} \geq n\}$
At-most number restriction	$\leq n R$	$(\leq n R)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \{b \in \Delta^{\mathcal{I}} \mid (a, b) \in (R)^{\mathcal{I}}\} \leq n\}$
Exact number restriction	$= n R$	$(= n R)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \{b \in \Delta^{\mathcal{I}} \mid (a, b) \in (R)^{\mathcal{I}}\} = n\}$

Name	Syntax	Semantics
Qualified at-least number restriction	$\geq n R.C$	$(\geq n R.C)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \{b \in \Delta^{\mathcal{I}} \mid (a, b) \in (R)^{\mathcal{I}} \wedge b \in (C)^{\mathcal{I}}\} \geq n\}$
Qualified at-most number restriction	$\leq n R.C$	$(\leq n R.C)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \{b \in \Delta^{\mathcal{I}} \mid (a, b) \in (R)^{\mathcal{I}} \wedge b \in (C)^{\mathcal{I}}\} \leq n\}$
Qualified exact number restriction	$= n R.C$	$(= n R.C)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \{b \in \Delta^{\mathcal{I}} \mid (a, b) \in (R)^{\mathcal{I}} \wedge b \in (C)^{\mathcal{I}}\} = n\}$
Role-value map inclusion	$R \subseteq S$	$(R \subseteq S)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \forall b. (a, b) \in (R)^{\mathcal{I}} \Rightarrow (a, b) \in (S)^{\mathcal{I}}\}$
Agreement	$R \doteq S$	$(R \doteq S)^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \forall b. (a, b) \in (R)^{\mathcal{I}} \Leftrightarrow (a, b) \in (S)^{\mathcal{I}}\}$
Role fillers	$R : \{a_1, \dots, a_k\}$	$(R : \{a_1, \dots, a_k\})^{\mathcal{I}} = \{a \in \Delta^{\mathcal{I}} \mid \{(a_1)^{\mathcal{I}}, \dots, (a_k)^{\mathcal{I}}\} \subseteq \{b \in \Delta^{\mathcal{I}} \mid (a, b) \in (R)^{\mathcal{I}}\}\}$
One-of	$\{a_1, \dots, a_k\}$	$(\{a_1, \dots, a_k\})^{\mathcal{I}} = \{(a_1)^{\mathcal{I}}, \dots, (a_k)^{\mathcal{I}}\}$
Nominal	$\{a\}$	$(\{a\})^{\mathcal{I}} = \{(a)^{\mathcal{I}}\}$

Here $n \in \mathbb{N}$. In role fillers and one-of constructors, the listed object names are interpreted as individual names in $\mathcal{I}$.

3 Concrete Syntax for Concept Constructors

Name	Concrete syntax	Abstract syntax
Top	<code>c_top</code>	$\top$
Bottom	<code>c_bot</code>	$\perp$
Atomic state concept	<code>(c_atomic_state "p")</code>	P
Positive atomic goal concept	<code>(c_atomic_goal "p" true)</code>	G^+
Negative atomic goal concept	<code>(c_atomic_goal "p" false)</code>	G^-
Intersection	<code>(c_and C1 ... Cn)</code>	$C_1 \sqcap \dots \sqcap C_n$
Union	<code>(c_or C1 ... Cn)</code>	$C_1 \sqcup \dots \sqcup C_n$
Negation	<code>(c_not C)</code>	$\neg C$
Value restriction	<code>(c_all R C)</code>	$\forall R.C$
Existential quantification	<code>(c_some R C)</code>	$\exists R.C$
At-least number restriction	<code>(c_at_least n R)</code>	$\geq n R$
At-most number restriction	<code>(c_at_most n R)</code>	$\leq n R$
Exact number restriction	<code>(c_exactly n R)</code>	$= n R$
Qualified at-least restriction	<code>(c_at_least n R C)</code>	$\geq n R.C$
Qualified at-most restriction	<code>(c_at_most n R C)</code>	$\leq n R.C$
Qualified exact restriction	<code>(c_exactly n R C)</code>	$= n R.C$
Same-as / agreement	<code>(c_same_as R1 R2)</code>	$R_1 \doteq R_2$
Role-value map	<code>(c_subset R1 R2)</code>	$R_1 \subseteq R_2$
Role fillers	<code>(c_fillers R a1 ... an)</code>	$R : \{a_1, \dots, a_n\}$

Name	Concrete syntax	Abstract syntax
One-of	<code>(c_one_of a1 ... an)</code>	$\{a_1, \dots, a_n\}$
Nominal	<code>(c_nominal a)</code>	$\{a\}$

4 Role Constructors

Name	Syntax	Semantics
Universal role	U	$(U)^{\mathcal{I}} = \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$
Atomic state role	P	$(P)^{\mathcal{I}} = \{(a, b) \in \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \mid P(a, b) \in s\}$
Positive atomic goal role	G^+	$(G^+)^{\mathcal{I}} = \{(a, b) \in \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \mid G(a, b) \in \gamma\}$
Negative atomic goal role	G^-	$(G^-)^{\mathcal{I}} = \{(a, b) \in \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \mid \neg G(a, b) \in \gamma\}$
Intersection	$R \sqcap S$	$(R \sqcap S)^{\mathcal{I}} = (R)^{\mathcal{I}} \cap (S)^{\mathcal{I}}$
Union	$R \sqcup S$	$(R \sqcup S)^{\mathcal{I}} = (R)^{\mathcal{I}} \cup (S)^{\mathcal{I}}$
Complement	$\neg R$	$(\neg R)^{\mathcal{I}} = (\Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}) \setminus (R)^{\mathcal{I}}$
Inverse	R^-	$(R^-)^{\mathcal{I}} = \{(b, a) \in \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \mid (a, b) \in (R)^{\mathcal{I}}\}$
Composition	$R \circ S$	$(R \circ S)^{\mathcal{I}} = \{(a, c) \mid \exists b. (a, b) \in (R)^{\mathcal{I}} \wedge (b, c) \in (S)^{\mathcal{I}}\}$
Transitive closure	R^+	$(R^+)^{\mathcal{I}} = \bigcup_{n \geq 1} ((R)^{\mathcal{I}})^n$
Reflexive-transitive closure	R^*	$(R^*)^{\mathcal{I}} = \bigcup_{n \geq 0} ((R)^{\mathcal{I}})^n$
Role restriction	$R C$	$(R C)^{\mathcal{I}} = (R)^{\mathcal{I}} \cap (\Delta^{\mathcal{I}} \times (C)^{\mathcal{I}})$
Identity	$\text{id}(C)$	$(\text{id}(C))^{\mathcal{I}} = \{(d, d) \mid d \in (C)^{\mathcal{I}}\}$

The iterated composition $((R)^{\mathcal{I}})^n$ is defined by

$$((R)^{\mathcal{I}})^0 = \{(d, d) \mid d \in \Delta^{\mathcal{I}}\}, \quad ((R)^{\mathcal{I}})^{n+1} = ((R)^{\mathcal{I}})^n \circ (R)^{\mathcal{I}}.$$

5 Concrete Syntax for Role Constructors

Name	Concrete syntax	Abstract syntax
Universal role	<code>r_universal</code>	U
Atomic state role	<code>(r_atomic_state "p")</code>	P
Positive atomic goal role	<code>(r_atomic_goal "p" true)</code>	G^+

Name	Concrete syntax	Abstract syntax
Negative atomic goal role	(r_atomic_goal "p" false)	G^-
Intersection	(r_and R1 ... Rn)	$R_1 \sqcap \dots \sqcap R_n$
Union	(r_or R1 ... Rn)	$R_1 \sqcup \dots \sqcup R_n$
Complement	(r_complement R)	$\neg R$
Inverse	(r_inverse R)	R^-
Composition	(r_composition R1 ... Rn)	$R_1 \circ \dots \circ R_n$
Transitive closure	(r_transitive_closure R)	R^+
Reflexive-transitive closure	(r_reflexive_transitive_closure R)	R^*
Role restriction	(r_restriction R C)	$R C$
Identity	(r_identity C)	$\text{id}(C)$

6 Boolean Constructors

Name	Syntax	Semantics
Positive atomic state predicate	$\text{holds}(p)$	true iff there exists an atom $p() \in s$
Negative atomic state predicate	$\neg \text{holds}(p)$	true iff no current atom with predicate p is true in the state
Positive atomic goal predicate	$\text{goal}(p)$	true iff $p() \in \gamma$
Negative atomic goal predicate	$\neg \text{goal}(p)$	true iff $\neg p() \in \gamma$
Non-empty concept	$C \neq \emptyset$	true iff $(C)^{\mathcal{I}} \neq \emptyset$
Non-empty role	$R \neq \emptyset$	true iff $(R)^{\mathcal{I}} \neq \emptyset$

7 Concrete Syntax for Boolean Constructors

Name	Concrete syntax	Abstract syntax
Positive atomic state predicate	(b_atomic_state "p" true)	$\text{holds}(p)$
Negative atomic state predicate	(b_atomic_state "p" false)	$\neg \text{holds}(p)$
Positive atomic goal predicate	(b_atomic_goal "p" true)	$\text{goal}(p)$
Negative atomic goal predicate	(b_atomic_goal "p" false)	$\neg \text{goal}(p)$
Non-empty concept	(b_nonempty C)	$C \neq \emptyset$
Non-empty role	(b_nonempty R)	$R \neq \emptyset$

8 Numerical Constructors

Name	Syntax	Semantics
Concept count	$ C $	$ (C)^{\mathcal{I}} $
Role count	$ R $	$ (R)^{\mathcal{I}} $
Role distance	$d_R(C, D)$	$\min\{n \in \mathbb{N} \mid (C)^{\mathcal{I}} \times (D)^{\mathcal{I}} \cap ((R)^{\mathcal{I}})^n \neq \emptyset\}$, or ∞ if empty

9 Concrete Syntax for Numerical Constructors

Name	Concrete syntax	Abstract syntax
Concept count	<code>(n_count C)</code>	$ C $
Role count	<code>(n_count R)</code>	$ R $
Role distance	<code>(n_distance C R D)</code>	$d_R(C, D)$

10 Relational Query Expressions

Queries are operands of features in the base, extended, and unsolvability families.

Fix the state, goal, and register context, with finite universe $\Delta = \Delta^{\mathcal{I}}$. A query Q has an ordered schema and denotation

$$\text{cols}(Q) = \bar{u} = (u_1, \dots, u_k), \quad k \geq 0, \quad u_i \neq u_j \ (1 \leq i < j \leq k), \quad (Q)^{\mathcal{I}} \subseteq \Delta^k.$$

For $t = (d_1, \dots, d_k) \in \Delta^k$ and a list $\bar{v} = (v_1, \dots, v_m)$ of labels from $\bar{u}$, $m \geq 0$, define

$$t_{\bar{u}}[u_i] = d_i \ (1 \leq i \leq k), \quad t_{\bar{u}}[\bar{v}] = (t_{\bar{u}}[v_1], \dots, t_{\bar{u}}[v_m]).$$

Write $\bar{x} = (x_1, \dots, x_m)$, $\bar{y} = (y_1, \dots, y_k)$, $\bar{u}_i = \text{cols}(Q_i)$, and $\bar{w} = \bar{u}_1 \cdot (\bar{u}_2 \setminus \bar{u}_1)$. Here $\cdot$ concatenates lists; list difference deletes matching labels without reordering. The semantics column gives the denotation of each expression; $\text{arity}(H) = k$.

Name	Syntax	Semantics
Atomic state query	$\rho_{\bar{x}}(H_s)$	$\{\bar{d} \in \Delta^k \mid H(\bar{d}) \in s\}$
Positive atomic goal query	$\rho_{\bar{x}}(H_g^+)$	$\{\bar{d} \in \Delta^k \mid H(\bar{d}) \in \gamma\}$
Negative atomic goal query	$\rho_{\bar{x}}(H_g^-)$	$\{\bar{d} \in \Delta^k \mid \neg H(\bar{d}) \in \gamma\}$
Concept lift	$\rho_{(x)}(C)$	$\{(d) \mid d \in (C)^{\mathcal{I}}\}$

Name	Syntax	Semantics
Role lift	$\rho_{(x,y)}(R)$	$(R)^{\mathcal{I}}$
Natural join	$Q_1 \bowtie Q_2$	$\{t \in \Delta^{ \bar{w} } \mid t_{\bar{w}}[\bar{u}_1] \in (Q_1)^{\mathcal{I}} \wedge t_{\bar{w}}[\bar{u}_2] \in (Q_2)^{\mathcal{I}}\}$
Projection	$\pi_{\bar{x}}(Q)$	$\{t_{\bar{u}}[\bar{x}] \mid t \in (Q)^{\mathcal{I}}\}$
Column renaming	$\rho_{\bar{y}}(Q)$	$(Q)^{\mathcal{I}}$
Equality selection	$\sigma_{x=y}(Q)$	$\{t \in (Q)^{\mathcal{I}} \mid t_{\bar{u}}[x] = t_{\bar{u}}[y]\}$
Value selection	$\sigma_{x=a}(Q)$	$\{t \in (Q)^{\mathcal{I}} \mid t_{\bar{u}}[x] = (a)^{\mathcal{I}}\}$
Union	$Q_1 \cup Q_2$	$(Q_1)^{\mathcal{I}} \cup (Q_2)^{\mathcal{I}}$
Difference	$Q_1 \setminus Q_2$	$(Q_1)^{\mathcal{I}} \setminus (Q_2)^{\mathcal{I}}$
Concept projection	$\pi_x^C(Q)$	$\{t_{\bar{u}}[x] \mid t \in (Q)^{\mathcal{I}}\}, \quad x \in \{u_1, \dots, u_k\}$
Role projection	$\pi_{x,y}^R(Q)$	$\{(t_{\bar{u}}[x], t_{\bar{u}}[y]) \mid t \in (Q)^{\mathcal{I}}\}, \quad x, y \in \{u_1, \dots, u_k\}, \quad x \neq y$
Non-empty query	$Q \neq \emptyset$	true iff $(Q)^{\mathcal{I}} \neq \emptyset$
Query count	$ Q $	$ (Q)^{\mathcal{I}} $

Schemas and side conditions. Column labels are unquoted identifiers matching `[A-Za-z-][A-Za-z0-9-]*`; every schema has distinct labels.

$$\begin{aligned}
\text{cols}(\rho_{\bar{x}}(H_s)) &= \text{cols}(\rho_{\bar{x}}(H_g^{\pm})) = \bar{x}, & |\bar{x}| &= \text{arity}(H), \\
\text{cols}(\rho_{(x)}(C)) &= (x), & \text{cols}(\rho_{(x,y)}(R)) &= (x, y), \quad x \neq y, \\
\text{cols}(Q_1 \bowtie Q_2) &= \bar{w}, \\
\text{cols}(\pi_{\bar{x}}(Q)) &= \bar{x}, & \{x_1, \dots, x_m\} &\subseteq \{u_1, \dots, u_k\}, \\
\text{cols}(\rho_{\bar{y}}(Q)) &= \bar{y}, & |\bar{y}| &= |\bar{u}|, \\
\text{cols}(\sigma_{x=y}(Q)) &= \bar{u}, & x, y &\in \{u_1, \dots, u_k\}, \\
\text{cols}(\sigma_{x=a}(Q)) &= \bar{u}, & x &\in \{u_1, \dots, u_k\}, \quad a \text{ a declared domain constant}, \\
\text{cols}(Q_1 \cup Q_2) &= \text{cols}(Q_1 \setminus Q_2) = \bar{u}_1, & \bar{u}_1 &= \bar{u}_2.
\end{aligned}$$

For the empty schema,

$$\Delta^0 = \{()\}, \quad t_{\bar{u}}[()] = (), \quad (\pi_{()}(Q))^{\mathcal{I}} = \begin{cases} \{()\}, & (Q)^{\mathcal{I}} \neq \emptyset, \\ \emptyset, & (Q)^{\mathcal{I}} = \emptyset. \end{cases}$$

11 Concrete Syntax for Relational Query Constructors

Name	Concrete syntax	Abstract Syntax
Atomic state query	(q_atomic_state "H" (x1 ... xk))	$\rho_{\bar{x}}(H_s)$
Positive atomic goal query	(q_atomic_goal "H" true (x1 ... xk))	$\rho_{\bar{x}}(H_g^+)$
Negative atomic goal query	(q_atomic_goal "H" false (x1 ... xk))	$\rho_{\bar{x}}(H_g^-)$
Concept lift	(q_concept x C)	$\rho_{(x)}(C)$
Role lift	(q_role (x y) R)	$\rho_{(x,y)}(R)$
Natural join	(q_join Q1 Q2)	$Q_1 \bowtie Q_2$
Projection	(q_project (x1 ... xm) Q)	$\pi_{\bar{x}}(Q)$
Column renaming	(q_rename (y1 ... yk) Q)	$\rho_{\bar{y}}(Q)$
Equality selection	(q_select_equal x y Q)	$\sigma_{x=y}(Q)$
Value selection	(q_select_value x "a" Q)	$\sigma_{x=a}(Q)$
Union	(q_union Q1 Q2)	$Q_1 \cup Q_2$
Difference	(q_difference Q1 Q2)	$Q_1 \setminus Q_2$
Concept projection	(c_project x Q)	$\pi_x^C(Q)$
Role projection	(r_project x y Q)	$\pi_{x,y}^R(Q)$
Non-empty query	(b_nonempty Q)	$Q \neq \emptyset$
Query count	(n_count Q)	$ Q $

For the displayed constructors, syntactic complexity is

$$\text{size}(\text{op}(E_1, \dots, E_m; \alpha)) = 1 + \sum_{i=1}^m \text{size}(E_i),$$

where E_i are expression operands and α collects non-expression arguments.

12 Unsolvability Family Constructors

The unsolvability family (**uns**) extends the base family with comparison operators and constants. A comparison takes two operands of the same category—either two Booleans or two Numericals—and yields a Boolean. Boolean operands are interpreted numerically as 0 (false) or 1 (true), so the same relational operators apply uniformly to both operand categories. We write $(e)^I$ for the numeric value of an operand e (a count/distance for Numericals, or 0/1 for Booleans).

Name	Syntax	Semantics
Equal	$e_1 = e_2$	true iff $(e_1)^I = (e_2)^I$
Not equal	$e_1 \neq e_2$	true iff $(e_1)^I \neq (e_2)^I$
Less than	$e_1 < e_2$	true iff $(e_1)^I < (e_2)^I$
Less or equal	$e_1 \leq e_2$	true iff $(e_1)^I \leq (e_2)^I$
Greater than	$e_1 > e_2$	true iff $(e_1)^I > (e_2)^I$
Greater or equal	$e_1 \geq e_2$	true iff $(e_1)^I \geq (e_2)^I$
Boolean constant	$b \in \{\text{true}, \text{false}\}$	the constant truth value b
Numerical constant	$k \in \mathbb{N}$	the constant nonnegative integer k

13 Concrete Syntax for Unsolvability Family Constructors

Name	Concrete syntax	Abstract syntax
Boolean equal	(b_eq B1 B2)	$B_1 = B_2$
Boolean not equal	(b_neq B1 B2)	$B_1 \neq B_2$
Boolean less than	(b_lt B1 B2)	$B_1 < B_2$
Boolean less or equal	(b_le B1 B2)	$B_1 \leq B_2$
Boolean greater than	(b_gt B1 B2)	$B_1 > B_2$
Boolean greater or equal	(b_ge B1 B2)	$B_1 \geq B_2$
Numerical equal	(n_eq N1 N2)	$N_1 = N_2$
Numerical not equal	(n_neq N1 N2)	$N_1 \neq N_2$
Numerical less than	(n_lt N1 N2)	$N_1 < N_2$
Numerical less or equal	(n_le N1 N2)	$N_1 \leq N_2$
Numerical greater than	(n_gt N1 N2)	$N_1 > N_2$
Numerical greater or equal	(n_ge N1 N2)	$N_1 \geq N_2$
Boolean constant	(b_const true) / (b_const false)	true / false
Numerical constant	(n_const k)	k

Source

This document summarizes Appendix 1, Section A1.2 of *The Description Logic Handbook*, edited by Franz Baader, Deborah L. McGuinness, Daniele Nardi, and Peter F. Patel-Schneider.